Lab: Computational Phenotyping and Individual Differences

Learning Goal: Design and validate computational phenotyping studies.

Part 1: Parameter Recovery Design

Exercise: For a probabilistic reversal learning task with an Active Inference POMDP model, design a parameter recovery study:

Model parameters to recover:

1. α (learning rate): How quickly beliefs about reward probabilities update
2. ω (prior volatility): How much the agent expects the environment to change
3. β (policy precision): How deterministically actions follow the softmax policy
4. C_reward (preference strength): How strongly the agent prefers reward vs. no-reward

Recovery procedure:

1. Choose 100 parameter combinations spanning the realistic range
2. For each, simulate 200 trials of behavior
3. Fit the model to each simulated dataset
4. Plot recovered vs. true parameters

Expected results table:

Parameter	Range	Recovery r^2	Common Issues
α	[0.1, 0.9]	Should be >0.85	May be confounded with ω
ω	[0.01, 0.5]	Should be >0.80	Boundary effects near 0
β	[1, 10]	Should be >0.90	Well-identified by choice variability
C_reward	[1, 5]	Should be >0.85	Confounded with β if no explicit reward manipulation

What would you do if α and ω are poorly separated ($r^2 < 0.5$)?

Write your response here

Part 2: Model Fitting to Data

Learning Goal: Practice fitting Active Inference models to empirical data.

Exercise: You have behavioral data from a reversal learning task. A participant made 200 choices with outcomes. Key patterns:

- Trials 1-80: Chose option A consistently (A rewarded 80% of the time)
- Trials 81-100: Continued choosing A after reversal (slow to switch)
- Trials 101-200: Gradually shifted to B

What do these behavioral patterns suggest about the computational phenotype?

Observation	Parameter Implication
Consistent initial choice	Moderate β (confident in policy)
Slow reversal detection	Low ω (prior: environment is stable) + moderate α
Gradual shift	Not extreme in any parameter

Estimate: Is this a "stable learner" (low ω, moderate α) or a "confused learner" (low α, high β)?

Write your response here

Part 3: Clinical Application

> **Learning Goal:** Design study using computational phenotyping for clinical classification.

Exercise: Design a computational phenotyping study for anxiety disorders:

Hypothesis: Patients with generalized anxiety disorder (GAD) have elevated precision on threat-related prediction errors (ω_{threat} ↑) and exaggerated volatility estimates (heightened uncertainty about safety).

1. **Task**: Go/No-Go task with threat and safe cues. Shock (unconditioned stimulus) follows threat cues with 70% probability. Safe cues never paired with shock.
2. **Model**: Active Inference POMDP with parameters: threat precision (ω_{threat}), safety precision (ω_{safe}), learning rate (α), policy precision (β)
3. **Groups**: 30 GAD patients, 30 matched controls
4. **Analysis plan**: Fit model to each participant → PEB group comparison → identify parameters that differ between groups

5. **Expected results**: GAD patients show higher ω_{threat} (over-weighting threat PEs) and possibly lower ω_{safe} (under-weighting safety evidence)

What additional analyses would strengthen the clinical utility? (Normative modeling? Treatment prediction? Longitudinal tracking?)

Write your response here

Part 4: Normative Modeling

> **Learning Goal:** Apply normative modeling to identify individual deviations.

Exercise: Given a population of N=200 healthy controls, each with a 4-parameter computational phenotype (α, ω, β, C), construct a normative model:

1. Fit a multivariate Gaussian to the control group: $N(\mu_{control}, \Sigma_{control})$
2. For a new patient, compute the Mahalanobis distance from the population center
3. Identify which parameters deviate most from the normative distribution

Example: Patient X has parameters $\alpha=0.9$ (very high), $\omega=0.05$ (very low), $\beta=8$ (high), C=2 (moderate)

Is this profile within the normative range or pathological? Which parameter(s) are most deviant?

Write your response here

Part 5: Reflection

In 300 words, reflect: Computational phenotyping promises personalized psychiatry, but faces challenges: model misspecification, test-retest reliability, and the question of whether computational parameters truly capture stable traits or fluctuating states. How should these challenges be addressed?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Validation design	Parameter recovery
2	Data interpretation	Behavioral-to-computational mapping
3	Clinical study design	Computational phenotyping for psychiatry
4	Statistical modeling	Normative modeling
5	Critical evaluation	Practical challenges